

Glide SP docking of the PAH panel into the periplasmic binding protein PahP

Ramin · branch [tool/glide](#) · Schrödinger Suite 2026-1, Glide v11.0

Target PahP, 308 aa Ligands 7 of 7 docked Mode SP, defined box Input commit fad5b9b Validator exit 0 Run 2026-09-02

SUMMARY

All seven canonical ligands docked into the shared PahP pocket with no failures, every best pose landing 0.95–2.72 Å from the grid centre — the same site CB-Dock3 reaches by blind docking. The run is deterministic: repeating it from the identical grid returned identical scores, which matters because Glide exposes no random seed. The scientifically interesting result is negative. GlideScore separates the seven ligands by only 0.64 kcal/mol and places the glucose negative control second; component analysis shows this is a genuine property of the scoring function on a pocket that is only 62% hydrophobic, not a defect in the pipeline. Neither Glide nor the comparison tools reproduce the published experimental affinity order for this protein.

1 Question and approach

Does PahP bind our PAH panel, and how does Glide rank that panel relative to the other docking engines in the comparison? The receptor, the seven ligands and the search region are shared across all workspaces, so the only variable between them is the engine itself. This report covers the Glide arm end to end: receptor preparation, ligand preparation, receptor grid, docking, and the standardised result file the comparison consumes.

2 Inputs, exactly as frozen

Receptor

[targets/PahP/structure.pdb](#), characterised before anything was altered: 4931 atoms, 308 residues, a single contiguous chain A, hydrogens already present from OpenMM, and no HETATM records, waters, metals or alternate locations. Critically, **no missing heavy atoms and no chain breaks** — almost everything a protein-preparation checklist normally repairs is simply absent here, which is why the protocol in Table 1 is deliberately short.

Ligands

[ligands/ligands.sdf](#) — seven molecules, all neutral with explicit hydrogens: six PAHs (128–252 Da, zero rotatable bonds) and glucose as a control. The longest, benzo[a]pyrene at 11.46 Å, sets the minimum legal grid dimension (Table 1).

Binding site

The canonical [binding_site.yaml](#) remains a placeholder with six null values and no [input-v1](#) tag. Because Glide has no blind-docking mode, a box is mandatory rather than optional, so one had to be sourced elsewhere.

3 Assumptions

- Receptor used exactly as frozen**, including its predicted N-terminal signal peptide (the file spans FASTA residues 10–317). The project plan calls for the mature domain, but every tool must dock an identical receptor, so this is escalated rather than fixed locally. The CB-Dock3 team raised the same point independently.
- Shared provisional box.** The grid uses the fpocket 4.2.2 Pocket 4 box that the AutoDock Vina team derived and proposed for the canonical file, centred at [70.83](#), [71.96](#), [59.18](#). This is a fairness decision, not a scoring one — see §4 for the three independent checks that it is the right site.
- Ligand chemistry untouched.** The six PAHs have no ionisable group and no tautomer at pH 7 ± 2, so ionisation and tautomer enumeration are chemically no-ops and were disabled. The one unavoidable change is that LigPrep always re-minimises geometry; chemistry, stereochemistry and protonation are unchanged.
- Rigid receptor, no seed.** Glide exposes no random seed, so [CANONICALIZE](#) is used as the documented substitute and the seed is recorded as [null](#) rather than invented.

STAGE	COMMAND AND KEY SETTINGS	OUTCOME AND RATIONALE
Receptor prep	prepwizard -rehtreat -disulfides -noepik -propka_pH 7.4 -f OPLS_2005	Hydrogens deleted and re-added under Schrödinger naming — the canonical ones come from OpenMM and their non-standard names break H-bond assignment. Restrained minimisation converged at 0.29 Å RMSD. Prime side-chain and loop filling deliberately omitted: nothing was missing, so it would only introduce model-dependent change.
Ligand prep	ligprep -i 0 -nt -s 1 -g	No ionisation, tautomeriser off (it is <i>on</i> by default), one stereoisomer, chirality read from the input 3D geometry. Seven structures in, seven out, all titles preserved.
Receptor grid	GRID_CENTER 70.83, 71.96, 59.18 · OUTERBOX 21.3, 19.6, 27.3 · INNERBOX 8	The inner box bounds only the ligand centroid; Glide requires outer ≥ inner + 11.46 Å on <i>every</i> axis, and the shortest axis is 19.6 Å, so 8 is the largest legal value. The script enforces this per axis and refuses to build a grid that would silently truncate benzo[a]pyrene.
Docking	PRECISION SP · confgen · POSES_PER_LIG 5 · POSTDOCK · CANONICALIZE · EPIK_PENALTIES False	XP is not licensed on this installation — PRECISION accepts only SP or HTVS. With Epik penalties and strain correction disabled, the reported docking score equals GlideScore exactly. WRITE_CSV and KEEPSKIPPED ensure no ligand can be silently dropped.
Extraction	collect_results.py → DOCKING_RESULT.json	Scores copied verbatim from Glide's native CSV; every canonical ligand receives a row, and a failure would carry Glide's own message. Validated against the project schema.

Table 1. Protocol as executed. Every flag was verified against the installed release via [prepwizard -h](#), [ligprep -long_help](#) and [glide -docking-keywords](#) rather than against external documentation.

4 Is this the right site?

Because the box was not canonical, it was checked three independent ways before the results were trusted. First, all seven CB-Dock3 *blind*-docking poses on PahP land 0.8–2.1 Å from this centre — an unrelated method converging on the same place. Second, a clearance scan of the canonical structure finds 24.8% of the inner box free of protein, with the nearest heavy atom 3.48 Å away: a real cavity, not filled space. Third, the site is lined by 16 residues, 62% hydrophobic, including five aromatics

#	LIGAND	GLIDESCORE (KCAL/MOL)	HEAVY ATOMS	LIGAND EFFICIENCY	POSE OFFSET FROM CENTRE	ROLE IN THE PANEL
1	anthracene	-6.114	14	-0.437	1.20 Å	specificity probe
2	glucose_NEGCTRL	-6.040	12	-0.503	0.95 Å	negative control
3	fluoranthene	-6.030	16	-0.377	2.72 Å	EQS Saar PAH
4	benzo_a_pyrene	-6.008	20	-0.300	1.96 Å	carcinogenic marker
5	phenanthrene	-5.938	14	-0.424	1.52 Å	degradation intermediate
6	pyrene	-5.883	16	-0.368	2.04 Å	primary target ligand
7	naphthalene	-5.471	10	-0.547	1.59 Å	smallest PAH

Table 2. Glide SP results for PahP, ranked by docking score (lower is better). Ligand efficiency is score divided by heavy-atom count. Pose offset is the distance from the best pose's heavy-atom centroid to the grid centre; the inner box permits up to ~4 Å by construction, so all seven poses are inside the intended site. Values copied verbatim from Glide's native CSV.

6 Interpretation

The negative control ranks second — a property of the score, not a fault

Every ligand occupies the same site, so this cannot be a placement artefact. The score components explain it directly: the PAHs score through lipophilic contact (−2.3 to −2.9) and van der Waals (−17 to −24) with *zero* hydrogen bonding and near-zero Coulomb, whereas glucose earns a weak lipophilic −1.0 but gains **H-bond −0.79 and Coulomb −8.87** against the pocket's polar face. A pocket that is only 62% hydrophobic genuinely offers a polyol something to bind.

The consequence is that glucose functions as a **plumbing and property control, not a specificity control**. It is mismatched to the PAHs on nearly every physicochemical axis — hydrogen-bond donors and acceptors, logP, sp³ character, stereocentres — so a score gap in either direction is confounded and carries little information about pocket selectivity.

How much of the ranking is ligand size?

Across the six PAHs, GlideScore correlates with heavy-atom count at **r = −0.68**. CB-Dock3, docking the same site, gives **r = −1.00**: its ranking is almost entirely explained by ligand size. Glide is therefore materially less size-driven, but it still does not predict affinity. Against the four ligands with published SPR measurements for this protein, the raw GlideScore order is *inverted* (Spearman ρ = −1.00), while size-normalised ligand

available for π-stacking (Phe123, Tyr271, Tyr274, Phe275, Phe306) alongside a polar face (Ser242, Ser243, Gln246, His250, Lys273).

5 Results

All seven ligands docked with no failures (Table 2). Every best pose sits inside the intended pocket, and total cost was 14.6 s CPU. Repeating the docking from the identical grid returned **identical scores for all seven ligands**, establishing that the pipeline is deterministic in practice despite the absence of a seed parameter.

efficiency is weakly positive (ρ = +0.20) and correctly ranks naphthalene first — matching the experimental finding that naphthalene binds most tightly. With n = 4 these coefficients are extremely noisy and no significance is claimed.

The defensible summary is that **neither engine reproduces the experimental affinity order**, which is the documented and expected behaviour of docking scoring functions. The comparison should therefore proceed on ranks, never on raw scores across tools.

7 Status and open items

- **The box is provisional.** Results are comparable in practice with the Vina and CB-Dock3 workspaces because all three searched the same region, but the box is not yet formally canonical. Freezing it into `binding_site.yaml` and tagging `input-v1` would make the comparison provably fair; every tool would then re-run from one definition.
- **Signal peptide.** Still present in the shared receptor; needs a central decision, not a per-workspace fix.
- **PahS not attempted.** PahP is the required minimum; CB-Dock3 has completed both targets.
- Reproduction instructions, including the determinism test, are in `REPRODUCE.md`; the full pipeline is drawn in `workflow.svg`. An earlier run used an incorrect box centre; that error and its diagnostics are recorded in `preprocessing/steps.md` so it is not repeated.